FROM THE

# EDITOR-IN-CHIEF

Technology has been changing the way we live since time immemorial. Whether it was the invention of the wheel, plumbing, electricity or other such conveniences, we take them so much for granted today. But all these were truly revolutionary at the time of their arrival. What technology does is make human beings more powerful in doing the tasks they do in everyday life. Since the invention of the computer—whether you take the room-sized version of 1945 or the first PC of 1973 as the original one—technological change has accelerated dramatically. Technology is like a fast-moving train that you must board sooner or later. Otherwise, you will be left on the platform and be doomed to irrelevance. With the arrival of the internet and the revolution in telecom, the world has been completely transformed in every aspect. The cell phone has become an integral part of most people's being. The two big issues for cell phone users is speed and latency. One is how quickly data is downloaded and the other is the time-lag between the initiation of a signal and its reception. With India on the threshold of the world of 5G, trouble on those two fronts will be a thing of the past. That is the promise of 5G. For example, it could enable you to download a three-hour movie like *RRR*, the southern blockbuster, to your device in a minute—which would take half an hour with even the best broadband service available at present in India. Also, because of ultra-low latency, call drops will be history.

Once this comes to pass, it will not just be in-home entertainment—5G will bring about a revolution across all aspects of our lives. Take three vital areas where 5G can effect a transformational change, benefitting millions at one go. Digital education, for one, a field of much promise where we were pushed into a fumbling start by the pandemic. Lightning-fast data transfer can unlock the world for our young minds. Or healthcare, where we can see point-of-care diagnostics—that is, right at the location of the patient—backed by connected ambulance services with real-time data transfer to the hospital. Or banking, where geospatial information systems can take us to simple, seamless and secure 'one-tap payments', eliminating layers of hassle-ridden steps.




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That's not all. In transportation and mobility, 5G can help electric vehicles identify and communicate with charging stations, and potentially reduce congestion on the highways by integrating transit systems such as FASTag for toll and entry tax. In agriculture, bots and drones can remotely monitor the health of crops, and sprinkle the right amount of fertilisers and pesticides. In manufacturing and industry, 5G would elevate automation efficiency to another level, reducing the carbon footprint. In governance and public safety, service delivery and citizen engagement could be improved with faster and safer digital verification.

Low latency will be the spark that can get a real fire going with the Internet of Things (IoT), the next revolution that has been prophesied for a while. There will be a massive cascading effect on applications and services. If a real-time network of reliable connections can be set up between cars in motion in a city and a centralised monitoring eye, imagine what traffic management can turn into. By 2023, connected cars will be "the largest installed base of 5G IoT endpoints" globally, reckons data company Statista. In India, even as we tailor the technology to our immediate needs, the possible futures are infinite.

As Executive Editor M.G. Arun records in our cover story, the potential market in India is enormous. The total mobile subscriber base stood at 1,143 million at the end of April 2022. About 70 per cent of that (some 800 million) is on 4G networks right now. This is the segment that will gradually migrate to 5G. But just like aeroplanes didn't kill trains, technological eras will overlap in telecom too. The millions on 2G and 3G networks—some peg them at about 300 million and 100 million respectively—will continue to exist. Together, that's a massive demographic in itself—even at conservative estimates, easily rivalling the entire population of the USA. But in the urban centres where it will be first available, things are expected to move at 5G speed. India is a glutton for data—the years 2017 to 2021 saw data consumption balloon dramatically at a compounded annual growth rate of 53 per cent. Technology will now stoke that appetite further.

However, the promise of all the goodies comes at a price. The telecom sector needs an estimated Rs 8 lakh crore investment in infrastructure: fibreoptic networks, which presently have just 30 per cent coverage, will need to be scaled up to 70 per cent. The number of telecom towers need to double from the present 700,000. The biggest barrier will be the availability of handsets: at 56 million right now, 5G-compatible handsets comprise only around 7 per cent of the total 800 million smartphones in India. To up that ratio, prices at the entry level will need to drop from the present Rs 15,000 to Rs 10,000. Experts say once the 5G era dawns for good, the prospects of higher volumes will inevitably entice manufacturers into easing their mark-ups.

Some 70 countries already have 5G, and another 15 have it partially. Altogether, some 493 operators in 150 countries and territories are investing in fifth-generation telecom—and many more in the US could potentially add to that. China is far ahead of us. Some 356 Chinese cities have 5G, and by 2021 it had already built some 961,000 base stations. The US has 296 cities hooked up. South Korea, the pioneer, has wall-to-wall coverage in 85 cities—though, curiously, only a quarter of its 4G subscriber base has shifted. Spain has made the expansion of 5G a cornerstone of its post-pandemic recovery. The Philippines—which gives India a real run for its money in BPO services—was an early bird. Wisely, it said yes to 5G in 2018-19 and now has 98 cities covered. Even Nepal may have turned out faster on the draw than India. Only last-minute glitches held up what would have been South Asia's first 5G trials. About time India dialled this number too. Couldn't be soon enough.

*(Aroon Purie)*